

Leveraging Large Language Models for Anomaly Detection of Cyber Advanced Persistent Threats

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Abstract

Cybersecurity defences often fail to spot Advanced Persistent Threats because these cyberattacks are built to stay invisible. Instead of relying on recognizable malware, attackers use ordinary system tools to hide their tracks. To uncover them, this project relies on Large Language Models to read computer logs like regular sentences and find strange activities. However, these AI models typically act as black boxes, meaning they do not explain their choices. This creates a massive hurdle for security teams who need solid proof before they can act on a warning. To address this, the project introduces a system that both finds cyber threats and clearly explains why it sounded the alarm. The framework highlights the exact words in a log that triggered the warning and provides mathematical proof to back it up. Testing used the DARPA Transparent Computing Engagement 3 dataset across five different operating systems. To keep the testing fair and realistic, the AI was trained strictly on 80% of normal logs so it could learn healthy system habits. It was then tested on the remaining 20% of normal logs mixed with 100% of the unseen attack logs, perfectly simulating a real-world attack.

In the first stage of the project, a model called `all-mpnet-base-v2` was paired with a basic autoencoder. This setup looked for anomalies by measuring errors, where a higher error meant a likely attack. The system performed incredibly well, easily outperforming older techniques like Isolation Forest and One-Class SVM. It pointed out the exact text fragments, such as strange file names or network details, that made the warning score go up.

The second stage involved a massive expansion. Testing covered 16 different AI models and 11 autoencoder setups across all five systems and five log types, leading to 4,400 separate experiments. Since real attacks are extremely rare and make up just 0.005% of the data, a strict scoring method called PR-AUC was used to ensure accurate results. To make the alerts understandable, a tool called SHAP was applied. The tests revealed that network logs gave the clearest warning signs. Switching to network logs alone turned one system's poor score into an excellent one. Smaller encoder models also processed structured logs much better than large language decoder models. Basic autoencoders outperformed complex probabilistic ones because they provided clearer and more reliable explanations.

The most successful combination used the THEIA system with network logs, the `gtr-t5-large` model, and a Denoising Autoencoder. This specific setup caught 98% of attacks while correctly recognizing normal traffic 94.23% of the time. Two other systems even managed to operate with zero false alarms. SHAP confirmed that the system consistently caught real threat indicators, flagging external IP addresses, unusual file locations, and unfamiliar hacking terms. This proves the model detects real attacker behaviour rather than random system noise. To confirm these findings, a removal test was conducted. The suspicious words highlighted by SHAP were deleted, and the logs were tested a second time. In every single case, the error score dropped dramatically. In one specific instance, removing just a few suspicious words caused the error to drop by 87.5%. This provides mathematical proof that those exact words triggered the alert. Ultimately, this project delivers a highly accurate and transparent threat detection system that gives security teams verified reasons for every alarm.

Two research publications were produced from this work. A conference paper establishing the foundational detection pipeline was successfully published and presented at the Multimedia University Engineering Conference (MECON) 2026. A journal paper detailing the full benchmark, architectural findings, and the novel explainable ablation study has been finalized and submitted to a Q1 high-impact journal.